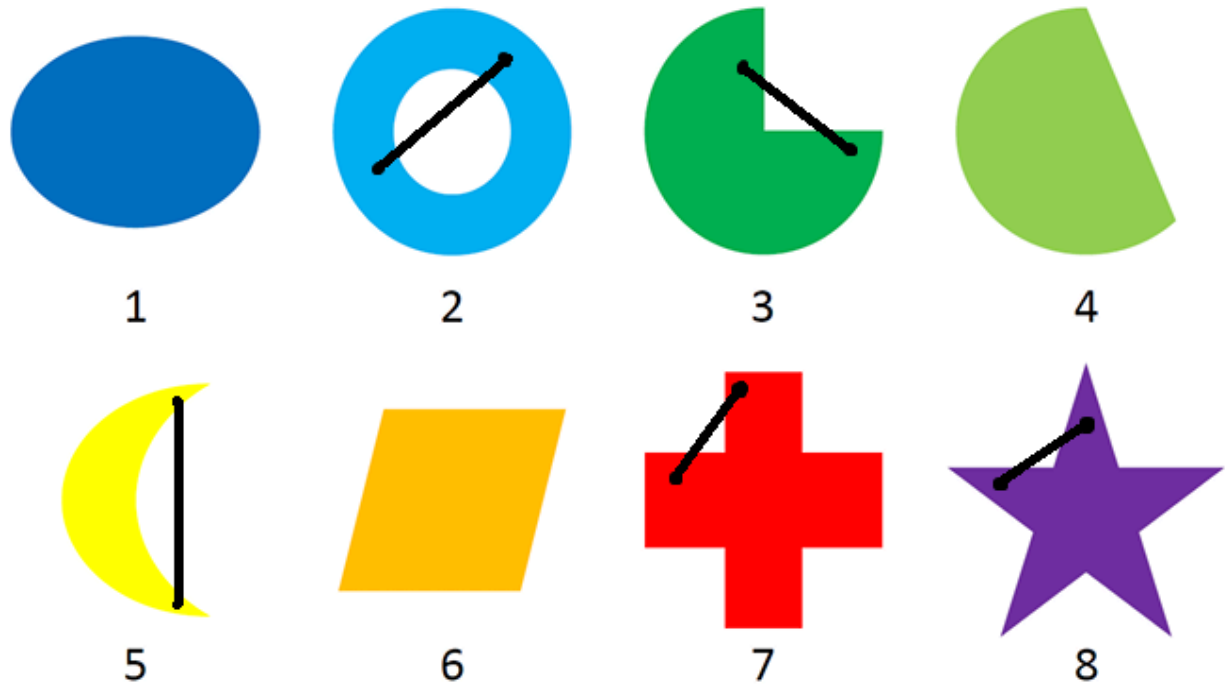
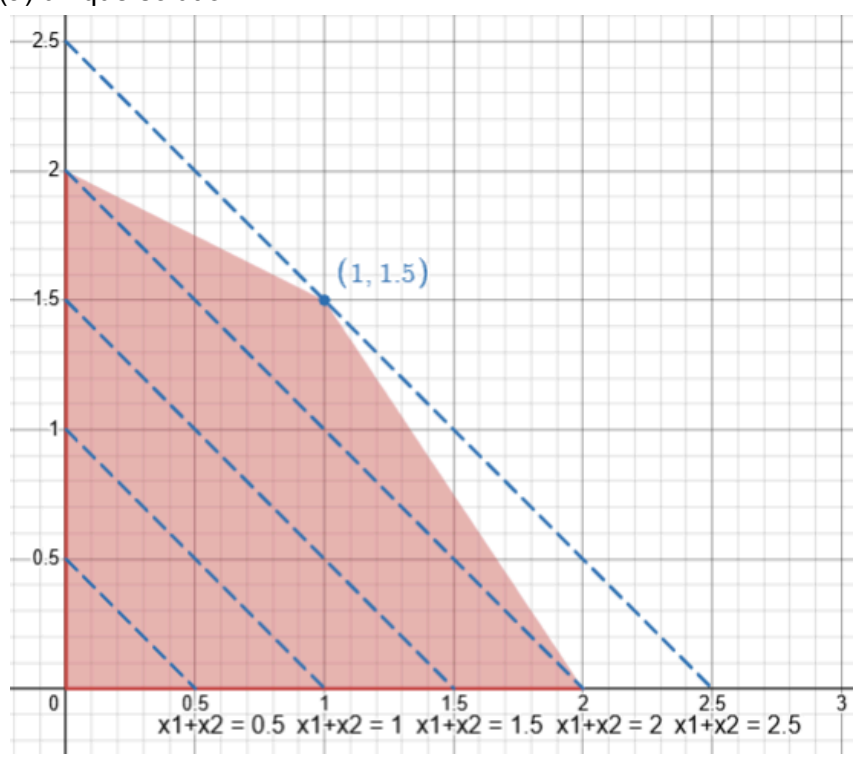
1. วิธีเช็ค convex set -> ถ้ามีวิธีเลือกจุดใน set มา 2 จุด แบบที่ลากเส้นตรงเชื่อมแล้วมันทะลุออกไปข้างนอก set แปลว่า set นั้นไม่ใช่ convex set



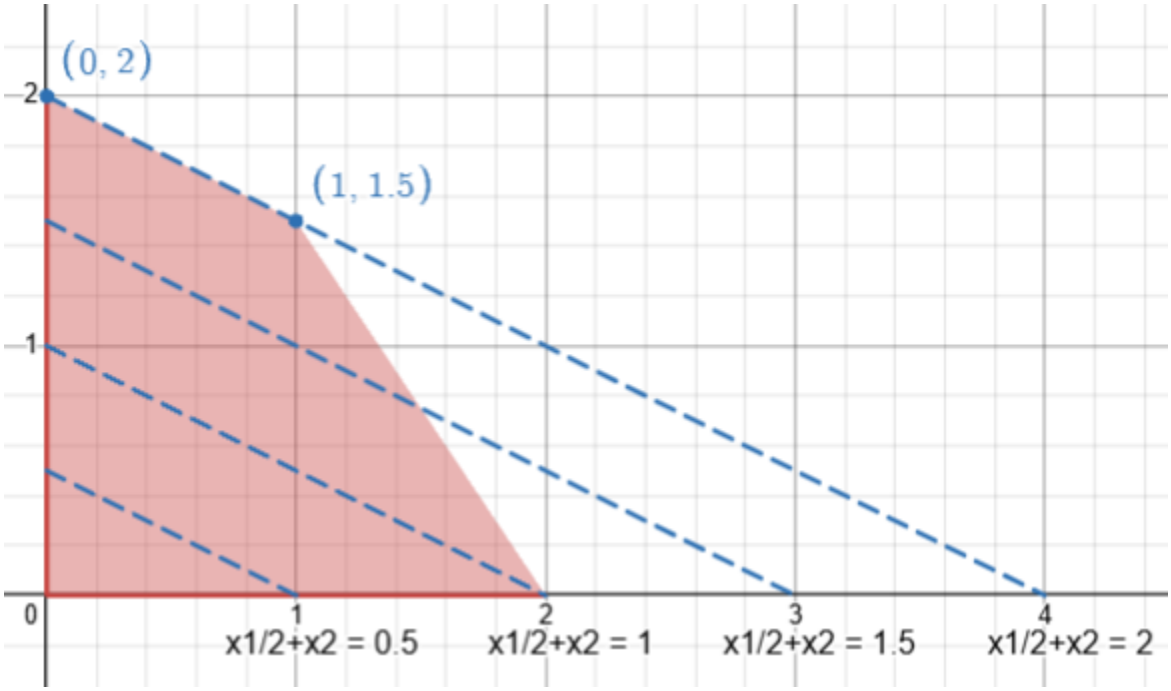
เหลือ 1 4 6 เป็น con2.vex set

2.

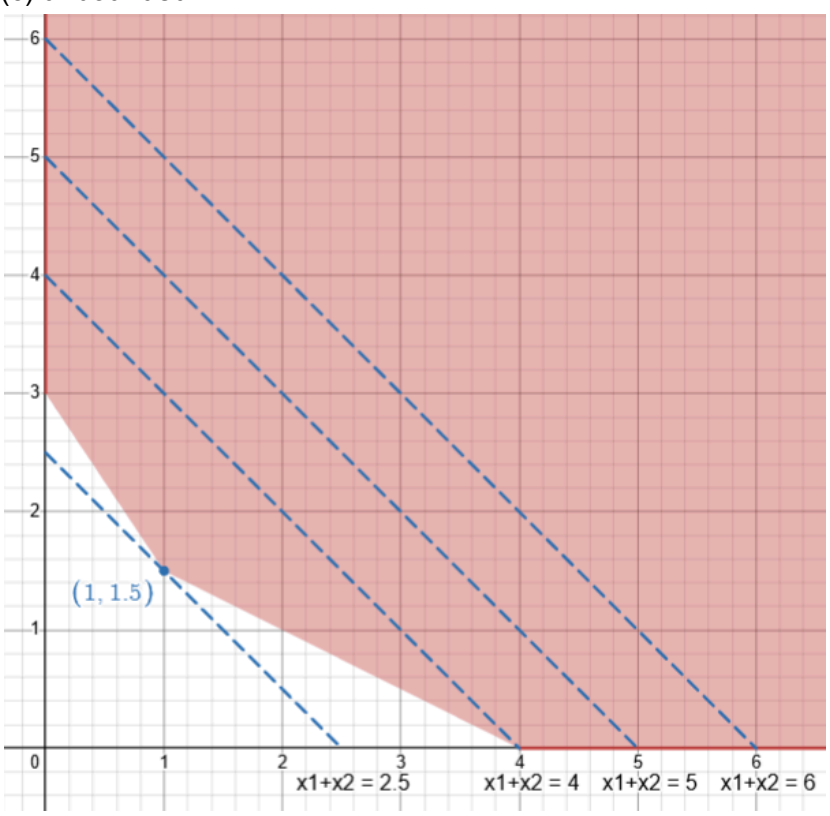
(a) unique solution



(b) infinite solutions



(c) unbounded



3. standard form:

$$
\begin{aligned}
\max\ z' = & -x_1 + x_2 \\
\text{subject to} \quad & x_1 + x_2 + s_1 = 3 \\
& x_1 + 3x_2 + s_2 = 4 \\
& 2x_1 - x_2 - e_1 = 1 \\
& x_1, x_2, s_1, s_2, e_1 \geq 0
\end{aligned}
$$

x_1	x_2	s_1	s_2	e_1	BFS?
0	0	3	4	-1	no
0	3	0	-2	-4	no
0	4/3	5/3	0	-7/3	no
0	-1	4	7	0	no
3	0	0	1	5	yes
4	0	-1	0	7	no
1/2	0	5/2	7/2	0	yes
5/2	1/2	0	0	7/2	yes
4/3	5/3	0	-7/3	0	no
1	1	1	0	0	yes

BFSs are $(x_1, x_2, s_1, s_2, e_1) = (\frac{1}{2}, 0, \frac{5}{2}, \frac{7}{2}, 0)$, $(1, 1, 1, 0, 0)$, $(\frac{5}{2}, \frac{1}{2}, 0, 0, \frac{7}{2})$, $(3, 0, 0, 1, 5)$.

4.

Problem 4. Let x_1, x_2, x_3 be the money (in million baht) awarded to a gold, a silver, and a bronze medalist, respectively. We have the following LP.

$$\begin{aligned} \min\ z = \ & x_1 + 2x_2 + 3x_3 \\ \text{subject to} \quad & 2x_1 + 6x_2 + 8x_3 = 60 \\ & x_1 - x_2 \geq 2 \\ & x_2 - x_3 \geq 1 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Standard form:

$$\begin{aligned} \max\ z' = & -x_1 - 2x_2 - 3x_3 \\ \text{subject to} \quad & 2x_1 + 6x_2 + 8x_3 = 60 \\ & x_1 - x_2 - e_1 = 2 \\ & x_2 - x_3 - e_2 = 1 \\ & x_1, x_2, x_3, e_1, e_2 \geq 0 \end{aligned}$$

x_1	x_2	x_3	e_1	e_2	BFS?	z'
0	0	15/2	-2	-17/2	no	
0	10	0	-12	9	no	
0	-2	9	0	-12	no	
0	34/7	27/7	-48/7	0	no	
30	0	0	28	-1	no	
2	0	7	0	-8	no	
34	0	-1	0	0	no	
9	7	0	0	6	yes	-23
27	1	0	24	0	yes	-29
6	4	3	0	0	yes	-23

BFSs are $(x_1, x_2, x_3, e_1, e_2) = (6, 4, 3, 0, 0)$, $(9, 7, 0, 0, 6)$, $(27, 1, 0, 24, 0)$.

The optimal BFS is not unique because both $(6, 4, 3, 0, 0)$ and $(9, 7, 0, 0, 6)$ yield the optimal value $z' = -23$, which is $z = 23$.